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Rage against the machine: Automation in the moral domain☆

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A B S T R A C T

The introduction of ever more capable autonomous systems is moving at a rapid pace. The technological progress will enable us to completely delegate to machines processes that were once a prerogative for humans. Progress in fields like autonomous driving promises huge benefits on both economical and ethical scales. Yet, there is little research that investigates the utilization of machines to perform tasks that are in the moral domain. This study explores whether subjects are willing to delegate tasks that affect third parties to machines as well as how this decision is evaluated by an impartial observer. We examined two possible factors that might coin attitudes regarding machine use—perceived utility of and trust in the automated device. We found that people are hesitant to delegate to a machine and that observers judge such delegations in relatively critical light. Neither perceived utility nor trust, however, can account for this pattern. Alternative explanations that we test in a post-experimental survey also do not find support. We may thus observe an aversion *per se* against machine use in the moral domain.

"I know I have made some very poor decisions recently, but I can give you my complete assurance that my work will be back to normal. I have still got the greatest enthusiasm and confidence in the mission. And I want to help you."
– HAL9000 (2001: A Space Odyssey)

1. Introduction

Due to the constant progress of automation over the past decades, we find ourselves ever and anon in a situation in which we have the possibility of employing an automated companion to help us take some work off our shoulders. In a perfect collaboration scenario the human operator delegates part of the work to her automated aid while she keeps an eye on its performance and takes back control whenever she sees fit. Yet, as technology progresses we (will) find ourselves in situations in which this dichotomy of work and supervision might crumble—even up to a point where human supervision during a task is neither needed nor wanted. The planned introduction of a technology that need and will not be monitored by human operators during its performance therefore poses new ethical challenges and questions. In the absence of a human operator who serves as an ultimately responsible moral agent, we have to address questions of responsibility and liability (Hevelke and Nida-Rümelin, 2014). Recently, the case of autonomous cars is gaining substantial interest.

Almost all car manufacturing firms have fostered the development of automated devices. While traditional car companies follow a step by step approach of adding pieces of automation to their latest models, such as "Active Lane Keeping Assist" systems, Google and Tesla are taking a disruptive approach that aims directly at the creation of a completely autonomous vehicle. The economic opportunities of autonomous driving are great. A Morgan Stanley report estimates a productivity gain of about $500 billion annually for the U.S. alone (Shanker et al., 2013). But there is also a moral case that can be made: Since most traffic accidents are due to human error (drunk driving, speeding, distraction, insufficient abilities) some estimate that the introduction of autonomous cars will decrease the number of traffic accidents by as much as 90% (Gao et al., 2014).

While a small literature on the moral case of autonomous driving exists, it mainly focuses on utilitarian benefits of the technology (Fagnant and Kockelmann, 2015) or deals with ethical decision-making in dilemma situations (see, e.g., Goodall, 2014; Gogoll and Müller, 2017). Little attention has been paid to possible empirical reservations that might influence the acceptance of the new technology. The delegation of a task that could carry severe consequences for a third party to an unmonitored machine might invoke popular resistance to the technology in cases of malfunction. This is of the utmost importance, since

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any form of public reservation regarding the introduction of new technology could impede the implementation of a technology that could be beneficial overall.

The relationship between human operators and automated devices has generated vast amounts of literature. The primary focus has been set on understanding this relationship. To our knowledge, however, the question of whether the delegation of tasks that affect a third party to an automated device is being welcomed or condemned has not received any attention. This may largely be due to the fact that the usual role of a human operator is to supervise and control an automated device that carries out a specific task. A typical example is the duties of the pilot of an airplane, which is, essentially, capable of flying on its own. The primary role of a human operator is therefore to supervise and—if need be—to intervene in case of automation failure or unforeseen scenarios that are not in the domain of the automated device. Consequently, a large part of the literature has investigated what factors influence the usage of an automated device.

Dzindolet et al. (2001) have created a framework of automation use indicating a variety of parameters that can be used to predict the use of automation in a human-computer "team". There is evidence that people who can opt in for automation use sometimes fear a loss of control when delegating to an automated device (Muir and Moray, 1996; Ray et al., 2008). This study, however, investigates the attitudes toward delegating tasks that affect a third party to a machine rather than to a human being, as opposed to the more general question of under which circumstances people are willing to relinquish control. The latter would refer to people's general propensity to delegate, as is also the case when people take a bus or taxi instead of driving themselves. To abstract from this issue, we wittingly forced subjects to give up control by delegating to either a machine agent or a human agent, thus keeping a loss of control constant between groups.

First, our study elicits attitudes toward machine use in the moral domain from the perspective of actors and observers: Do subjects prefer to delegate a task that affects a third party to a machine or a human? To what extent do subjects get blamed or praised for their delegation decision? Specifically, our first two hypotheses are as follows:

Hypothesis 1. People's delegation of a task that affects a third party to a human or to a machine is not balanced.

Hypothesis 2. Delegators are rewarded differently for delegating a task that affects a third party to a machine than for delegating it to a human.

In a second step, we investigate potential reasons for any negative or positive preference concerning machine use in the moral domain. Specifically, we test two factors that recur throughout the literature.

A major factor that could influence the decision of a subject to delegate to a machine is the "perceived utility" of an automated aid, which is defined as a comparison between the perceived reliability of an automated device and manual control (Dzindolet et al., 2002). If a subject judges that her ability exceeds that of an automated device, she usually does not allocate a task to the machine (Lee and Moray, 1994). This judgment might also be due to self-serving biases that see people overestimating their own abilities (Svenson, 1981) or their contribution to a joint task (Ross and Sicoly, 1979). Additionally, the perceived abilities of an automated aid might be influenced by a higher or lower salience of errors an automated device commits. There are controversial findings in cognitive psychology as to whether a violation of expectation (expectancy-incongruent information) is more readily remembered than decisions that are in line with prior anticipation (expectancy-congruent information) (Stangor and McMillan, 1992; Stangor and Ruble, 1989). While people initially tend to have high expectations of the performance of automation, humans may be judged according to a "errare humanum est" standard—decreasing the salience of an observed mistake made by a human delegatee due to a priced-in expectancy of errors. In Dzindolet et al. (2002) subjects chose to be paid according to their performance rather than that of their automated aids. This was even the case when they were informed that the automated device was far superior, stating salient errors of the automated device they perceived earlier to justify their decision. This is astonishing since an important factor in the decision to employ an automated device lies in the goal-oriented nature (Lee and See, 2004) of the task. Prima facie, a subject should be more likely to use automation if she rates the device's ability to successfully perform the delegated task positively (Davis, 1989), i.e., if the machine is seen as a reliable entity. We isolate the potential effect of machine-error salience by forcing subjects to relinquish control, thus abstracting from a self-serving bias. Our third hypothesis is as follows:

Hypothesis 3. Machine errors are perceived differently to human errors.

Another important factor that is known to influence the decision to delegate to an automated device is trust. The concept of trust has attracted a lot of attention regarding its influence on automation. While some researchers have seen trust between human agents and machines to be closely related to the traditional concept of trust between humans, others stress important differences regarding trust relationships between humans and machines (de Visser et al., 2012). Trust is a notoriously broad term but one characteristic that is commonly shared by most authors is a state of vulnerability the trustor has to be in. That is, a trust relationship requires the trustor's willingness to set herself in a vulnerable position by delegating responsibility to the trustee (Rousseau et al., 1998). Obviously, if the outcome of a delegation is completely determined and the process fully transparent, there is no need to incorporate trust. In this study, we use a simple trust game to isolate the mere aspect of trust, since it requires no capabilities on the trustee's side about which the trustor might have biased beliefs. The trust game only requires the trustee to reciprocate. It thus abstracts from the aspect of perceived utility discussed above, which is closely related to the specific task at hand. Finally, our fourth hypothesis is as follows:

Hypothesis 4. The level of trust toward machines and toward humans is different.

2. Experiment design

The experiment consisted of three parts: (1) the delegation and execution of a task that affected a third party, (2) a perception guess, and (3) a trust game. The aim of part 1 was to elicit attitudes toward machine use in the moral domain from the perspectives of actors and observers (Hypotheses 1 and 2). Part 2 was designed to test whether a given divergence in judgments towards humans and machines could stem from systematically different perceptions of the errors committed by humans versus machines (Hypothesis 3). Part 3 was designed to test whether different levels of trust in humans and machines could account for diverging judgments (Hypothesis 4).

Subjects received instructions for the experiment on screen. They were informed at the beginning that the experiment consisted of three independent parts and that they could earn money in each of these three parts. In the end of the experiment, one of these parts was selected at random and subjects were paid according to their respective payoff in this part. Prior to the experiment there were two preparatory sessions which provided us with the necessary data to calibrate machine performances and were also used to create perception tasks. We will first explain the three parts of the experiment and then provide some details on the preparatory sessions.

2.1. Part 1: Task affecting third party

Part 1 of the experiment consisted of the delegation of a calculation task to either another human or to a machine and in the subsequent solving of the task by human task-solvers and the machine. The

benevolent effort of the other human or the preprogrammed actions of the machine then determined the payoff of a third party.

For part 1 of the experiment, half of the subjects were randomly assigned the role of actors, and the other half, observers. One observer was randomly assigned to each actor. Each actor played all roles consecutively. First, actors as delegators had to delegate the calculation task either to another human or to a machine. Second, actors as task-solvers had to perform the task themselves. Third, actors as third parties were the recipient of the payoff created by the benevolent effort of another human task-solver or by the performance of a machine. The fact that the successful or unsuccessful performance of the task determined a third party's payoff made its solving, and—more importantly—its prior delegation, morally relevant.

Actors were first informed that they were randomly assigned to two other subjects in the lab, say X and Y. They were told that their own payoff would depend on the decision of Y and that their own decision would determine the payoff of X. The calculation task in part 1 of the experiment was then explained to actors. For the task each subject was confronted with a block of ten calculation exercises, each consisting of seven digits, lined up on the screen. The sum of the seven digits had to be entered in an input field. Finally, one line was selected at random. If the respective exercise was solved correctly, the third party received 70 ECU. Otherwise, the third party received nothing.

Before the actors made their delegation decision, we wanted them to form an impression about the relative capability of human task-solvers and the machine. Because we were interested in a potential systematic misperception of human and machine errors, we did not simply provide subjects with statistics on actual performances. Instead, all subjects were visually presented with past performances of 24 subjects from a preparatory session. They were also shown the corresponding performance of a preprogrammed algorithm (see section 2.4 for details).

Relative performances of humans and machine in the task were visualized on a split screen. The caption "human" and "machine" was shown on the respective half of the screen. In total, subjects were shown 240 (24 subjects solved 10 lines each) past solutions of humans and the corresponding machine performances. If a single exercise was solved correctly by the human subject or by the algorithm respectively, it appeared in white. Otherwise, it appeared in red.[1] Exercises solved by human and machine appeared alternately and one by one. Each exercise appeared for only 0.5 s making it extremely difficult to simply count the number of red lines. The side of the screen on which the performance of the machine was presented was randomized across subjects. In fact, subjects in the earlier preparatory sessions, and consequently also the tailored algorithm, solved about 20% of the lines incorrectly.

Once delegators had formed an impression of the performance of humans and the machine, they made their delegation decision. Note that every actor solved the calculation task in the task-solver's role for her recipient. Each actor did this without knowing whether her delegator had actually delegated the task to her or to a machine. This was done to prevent a general tendency to delegate to the machine to spare fellow subjects the work. The performance of a task-solver was only relevant for her recipient if the task-solver's delegator decided to delegate to her and not to a machine. Observers solved the calculation task as well, without any consequence for another subject, in order to give them an impression of the task.

Each actor was rewarded or punished for her delegation decision by her assigned observer. An observer could reduce the actor's initial endowment of 30 ECU by any integer amount, down to a minimum of zero

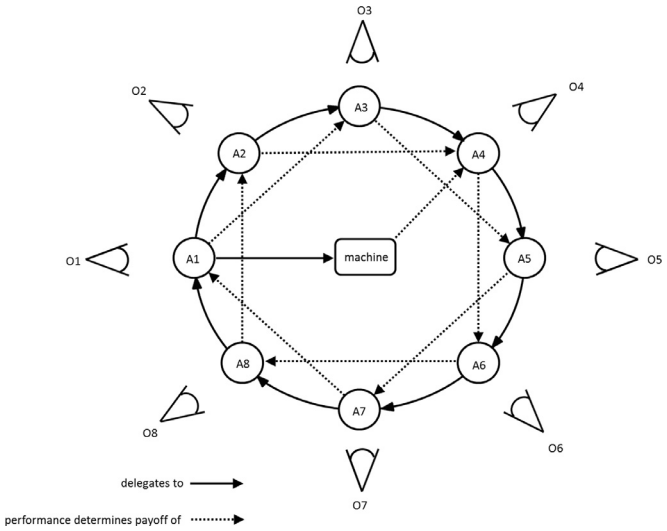


Fig. 1. Matching for delegation decision.

or increase it up to a maximum of 60 ECU without any influence on her own payoff. The observer could, of course, also leave the actor's endowment unaltered. Reward and punishment choices were elicited via the strategy method (Selten, 1967). This means that each observer made her reward or punishment choice conditional on the delegation decision, as well as its outcome. Thus, judgment was contingent upon whether the delegator had delegated to a human task-solver or to a machine *and* upon whether the randomly drawn exercise was solved correctly or not. An observer thus gave his full evaluation profile behind the veil.

For the first round, actors thus received their altered endowment, ranging from 0 to 60 ECU plus 70 ECU, if their task-solver had calculated the randomly drawn line correctly. Observers received a flat payment of 100 ECU for the first round.[2]

The dependencies between subjects and the matching procedure for part 1 of the experiment are illustrated in Fig. 1. Here, actors are denoted by the letter A, while observers are denoted by the letter O. Consider the case of A1. A1 delegates the calculation task to A2 or to a machine (solid arrows). A2's or the machine's performance in the calculation task then determines the payoff of A4 (dotted arrows).[3] In this constellation, A1 is the delegator, A2 is the task-solver, and A4 is the recipient. A1, however, is also a task-solver, because A8 delegates to him or to a machine. Finally, A1 is a recipient. His payoff depends on the calculation performance of A7, if A6 has decided to delegate the calculation task to A7. Otherwise, it depends on the machine's performance. As can be seen, the design made sure that there were no direct interdependencies between any actors in the experiment. Potential feelings of reciprocity were thus excluded.[4] Subjects were explicitly informed about this feature of the design. O1 rewarded or punished the delegation decision of A1.

In the example above, the task-solving performance of A2 only determined A4's payoff if A1 had actually delegated the decision to A2. Otherwise, the performance of the machine was relevant. In either case, one of the ten solved exercises was selected at random and the recipient

[1] Subjects were told the following: "To evaluate the performance of a person and a machine, you will subsequently see a comparison of the performance of a past run. One line is shown per column, each respectively calculated either by a person or a machine. If calculated correctly, the line will be displayed in white. If calculated incorrectly the line will be displayed in red."

[2] This equalized the observer's own payoff with the payoff of an as-yet unrewarded or unpunished actor who had received the 70 ECU from the randomly-drawn exercise solved successfully by the task-solver on whom he depended. This is the case because he was additionally equipped with an initial endowment of 30 ECU. Thus, we established a conservative measure of reward and punishment, since any alteration of actors' endowment by a generally inequality-averse observer would require good reasons.

[3] For reasons of visual clarity, delegation and payoff dependency between actors and machine in Fig. 1 are shown for A1 and A4 only, by way of an example.

[4] See Greiner and Levati (2005) regarding the issue of indirect reciprocity in small groups.

Table 1
Payoffs for accuracy of guess.

Deviation(%)	Payoff
≤ 20	70 ECU
≤ 40	40 ECU
≤ 60	20 ECU
> 60	0 ECU

received her earnings if it was solved correctly. If A1 delegated to the machine, the performance of the tailored algorithm determined the payoff to A4.

2.2. Part 2: Perception guess

For part 2, the role differentiation of subjects was abolished. Subjects were informed that they would soon be confronted with yet another visualization of actual previous performances of the known calculation task by humans and a machine.[5] Their task was then to guess the number of errors of either the humans or the machine as accurately as possible. When seeing the visualization, they did not yet know whether they would later be asked to guess the errors of the humans or of the machine. All subjects were shown the data of the 24 subjects (240 lines) from a second preparatory session (i.e., different data than used for visualization in part 1) and the performance of the tailored algorithm. As in part 1, the relative performance of humans and the machine was presented on a split screen. The side of the screen on which the performance of the machine was presented was randomized across subjects. Exercises solved correctly again appeared in white, while those solved incorrectly appeared in red. In order to prevent subjects from counting, each exercise was only shown for 0.3 s. The interval was even shorter than in part 1, because subjects were already used to this kind of visualization.

After the actual past performances were shown, subjects had to state how many of the 240 exercises shown had been solved incorrectly, i.e., how many errors had been made. Subjects' payoff for part 2 depended on the accuracy of their guess. Payoffs were calculated according to Table 1.

Half of subjects were randomly asked to guess humans' performance, while the other half was asked to guess the machine's performance. It was ensured that an equal number of actors and observers from part 1 of the experiment were distributed between both of these treatments. Furthermore, subjects who delegated to a human and those who delegated to a machine were also divided equally between the two treatments.

2.3. Part 3: Trust game

In part 3, subjects were randomly assigned to one of two treatments. In the Human Treatment, subjects played a standard trust game. Trustors were endowed with 50 ECU. They could transfer 0, 10, 20, 30, 40 or 50 ECU to the trustee. The sent amount was tripled and credited to the trustee. The trustee could then reciprocate any integer amount she wished. The Machine Treatment was identical to the Human Treatment except for the fact that the reciprocation decision was made by a machine agent on behalf of the trustee who had no chance to intervene. Before subjects were informed about the treatment to which they were assigned, the setup of both treatments was carefully explained to them.

In the Human Treatment, before subjects learned their role, they made their choice for the trust game via the strategy vector method (Selten, 1967) and submitted their full strategy profile for both roles. If a subject was ultimately assigned the role of a trustee, her reciprocation decision conditional on the amount actually transferred by the trustor was returned.

Because the trustee had no voice in the Machine Treatment, subjects only took a decision for the case of ending up in the role of the trustor.[6] The reciprocation decision of the machine was determined according to the actually previously submitted strategy profiles of the subjects in the preparatory sessions. The algorithm was programmed such that it picked one of all 48 reciprocation profiles submitted in the preparatory sessions at random, and applied the respective conditional choice to a trustor's actually chosen transfer.

Before subjects made their choices, they were given an impression of the reciprocation choices of humans and the machine on a split screen.[7] For each subject, the choices of the machine were shown on either the left or right side of the screen at random. For this purpose, subjects were shown the actual reciprocation profiles of all 48 subjects from both preparatory sessions. These choices were contrasted with the reciprocation profiles of the machine algorithm. Each profile consisted of five choices, i.e., the returned amount for each possible transfer. The five choices of a human and the machine profile selected at random appeared alternately and one by one in blocks. Each choice was shown for only 0.7 s.[8]

As in part 2, random assignment to the Human and Machine Treatment was contingent upon the subjects' role and delegation decisions from part 1. Thus, they were assigned in equal proportions to both treatments.

2.4. Preparatory sessions

The preparatory sessions were necessary for two reasons. First, they were needed in order to produce actual data from human task-solvers, which would later be presented to subjects in the experiment. Second, they were needed in order to tailor the machine's task-solving performance and decisions in the trust game to the performance and decisions of the humans. Keeping the de-facto performance of humans and machine constant allowed us to test for a potential systematic misperception of relative performances.

In the first part of the preparatory sessions, subjects processed the same calculation task as in the experiment. Also, each subject solved the task not for herself but for another subject with whom she was randomly matched. This receiver was paid according to the task-solvers performance. For this purpose, one of the ten exercises was selected at random and if this exercise was solved correctly, the receiver was given 70 ECU. Otherwise, she received nothing. It was ensured that no pair of subjects solved tasks for each other. So, the mechanism of the matching was the same as described in 2.1 in order to eliminate any potential feelings of reciprocity. Twenty-four subjects took part in each calibration session for the calculation task. Thus, 24 blocks of ten exercises were solved in each session.

[5] Subjects were told the following: "Now you will see the performance of humans and machines again. Please be aware of the fact that the data has been collected in a different past run than the performance that you saw in the first part."

[6] Subjects were told the following: "Every participant is able to send money to the participant assigned to him. This participant cannot decide how much money he wants to send back. This decision is made by a machine. You and the participant assigned to you decide simultaneously, but only one decision is going to be implemented. (…) The amount you transfer will be subtracted from your initial endowment. Subsequently, it will be tripled and send to the participant assigned to you. (…) Afterwards, the machine, deciding for the participant assigned to you, determines the amount of ECU that is returned to you. The participant assigned to you cannot influence the machine's decision. (…). As mentioned above, the participant assigned to you is also able to transfer money. The procedure is the same as already outlined above, meaning the returned amount is determined by your machine agent."

[7] Subjects were told the following: "To be able to form your personal expectations about how much will be sent back, you will be shown the return transfer of participants from an earlier session. To form an expectation about the return transfers of machines, you will see the decisions of the machine agent next to those of human participants."

[8] A choice was indicated by the returned amount for each possible transfer, e.g., "transfer: 30 → return:45."

The algorithm of the machine that solved the calculation task was programmed in such a way that it resembled the error distribution of the human subjects exactly.[9] So, for instance, if few subjects tended to make many errors, while many subjects made few errors, this was resembled by the algorithm: It made many mistakes in few of the 24 blocks and solved many blocks with few mistakes. The clustering of errors was important to equalize error distribution and account for risk preferences.[10]

Recall that past data on calculation performance was presented twice in the experiment, once before the delegation decision in part 1 and once before the perception guess in part 2. Therefore, two preparatory sessions were performed. We used the data from the first session for part 1 of the experiment, and the data from the second session for part 2.[11]

In the second part of the preparatory sessions, subjects were randomly rematched to new pairs and played a trust game with the same parameters as in the experiment. Using the strategy vector method, each subject gave a reciprocation profile for the case of ending up in the role of a trustee. A random draw then assigned the roles and payoffs were determined according to their own decision for that role and the decisions of their match. The collected 48 reciprocation profiles constituted the pool of data from which the machine agent in part 3 of the experiment randomly picked one and applied it to a trustor's chosen transfer.

Finally, one of the two parts of the preparatory sessions was selected at random and subjects were paid according to their payoff in this part.

3. Experiment results

The experiment took place in a major German university in September 2015. It was programmed in z-Tree (Fischbacher, 2007), subjects were recruited via ORSEE (Greiner et al., 2003). A total of 264 subjects participated in twelve sessions. Subjects received a show-up fee of € 4.00 and could earn additional money in the experiment. A session lasted about 45 min. and the average payment was € 10.38 (sd= € 3.45). Task-solvers solved on average 8.58 (sd = 2.34) of the ten exercises correctly. The conversion rate was 10 ECU = € 1.00.

First, we checked whether subjects preferred delegating a task that affects a third party to a human over delegating it to a machine. Overall, 132 subjects made a delegation decision. Ninety-seven of these subjects (73.48%) delegated to a human, 35 of them (26.52%) delegated to a machine. The fraction of subjects deciding to delegate to a machine is therefore significantly lower than half ($p < .001$, according to an Exact Binomial Test). This confirms our first hypothesis.

Result 1. Subjects preferred to delegate a task that affects a third party to a human than to a machine.

We now turn to analyze the observers' evaluation of a delegation to a machine as compared to a human. Remember that each observer evaluated both cases—the delegation to a human and to a machine—in a random order. Furthermore, he made choices contingent upon whether the respective task-solver had successfully solved the task or made an error. This means that each observer provided four choices.

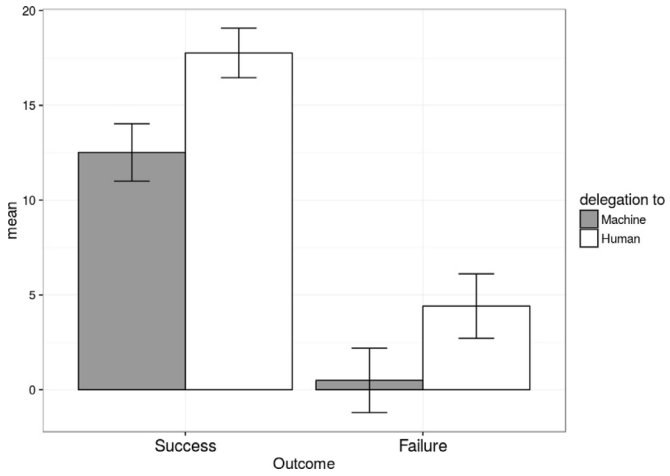


Fig. 2. Observers' rewarding of delegation to machines and to humans.

Observers' levels of rewarding delegators are illustrated in Fig. 2. If the respective task-solver was successful, observers rewarded delegations to a machine with an average of 12.52 ECU (sd= 17.38 ECU), while they rewarded delegations to humans with an average of 17.77 ECU (sd= 15.01 ECU). If the respective task-solver made an error, observers rewarded delegations to a machine with an average of 0.49 ECU (sd= 19.53 ECU), while they rewarded delegations to humans with an average of 4.42 ECU (sd= 19.53 ECU). Delegators to machines are thus evaluated significantly worse than delegators to humans regardless of whether the outcomes are successful or unsuccessful ($p < .001$ and $p = .002$, respectively, according to two-sided Wilcoxon signed-rank tests). This confirms our second hypothesis.

Result 2. Delegators were rewarded less for delegating a task that affects a third party to a machine than for delegating it to a human.

In the next two steps, we investigated whether the aversion to machine use in the moral domain identified is based on a lower "perceived utility" of the machine or on a general lack of trust in machines.

First, we compare the number of machine errors guessed by subjects to the number of human errors they guessed. Note that the number of actual errors, i.e., red-colored exercises, presented to subjects was the same for humans and the machine. Specifically, 50 of the 240 exercises shown to subjects on each side of the screen, which was split between human and machine, were shown in red. Subjects who were incentivized to guess the number of machine errors made an average guess of 58.11 (sd = 24.88), while those who were incentivized to guess the number of human errors made an average guess of 59.84 (sd = 24.43). This difference in guesses is insignificant ($p = .632$ according to a two-sided Mann–Whitney U-test). We thus reject our third hypothesis.

Result 3. Machine errors are not perceived significantly different from human errors.

Second, we tested whether the amount in the trust game transferred by trustors to a machine agent was lower than that sent to a human trustee. Those who were randomly matched with a machine agent sent an average amount of 30.83 ECU (sd = 16.67 ECU), while those matched with a human trustee sent an average of 33.64 ECU (sd= 15.78 ECU). The difference is insignificant ($p = .170$ according to a two-sided Mann-Whitney U-test).

One might suspect that this insignificance is only an aggregate phenomenon: It may result from the leveling of diverging levels of trust

[9] The algorithm was programmed such that it could not solve exercises in which the sum of numbers was higher than 34. The algorithm was fed with calculation data which led it to reproduce the historical error distribution from the calibration session precisely. Assume the first task-solver in a calibration session had made one mistake, while the second had made three mistakes, and so on. The algorithm was thus fed with an initial block of ten exercises in which one exercise added up to more than 34 and with a second block of ten exercises in which three exercises added up to more than 34. One of the 24 blocks resembling the performance of the task-solvers from the calibration study was randomly drawn to be decisive. The machine then actually calculated this block of ten exercises. Due to its inability to calculate the exercises adding up to more than 34 correctly, it made the same number of errors as the respective human task-solver.

[10] If the machine would have taken the average error rate of all 24 humans and "applied" it to each block, it would have caused a more uniform distribution of errors over the 24 blocks than the humans. In this case, a risk-averse subject might have preferred to delegate the task to a machine, because she feared a particularly weak fellow human subject more than she appreciated a particularly strong fellow human subject.

[11] The average number of lines solved correctly was 8.00 (sd = 2.10) in the first session and 7.92 (sd = 1.93) in the second session. They were thus very close to each other.

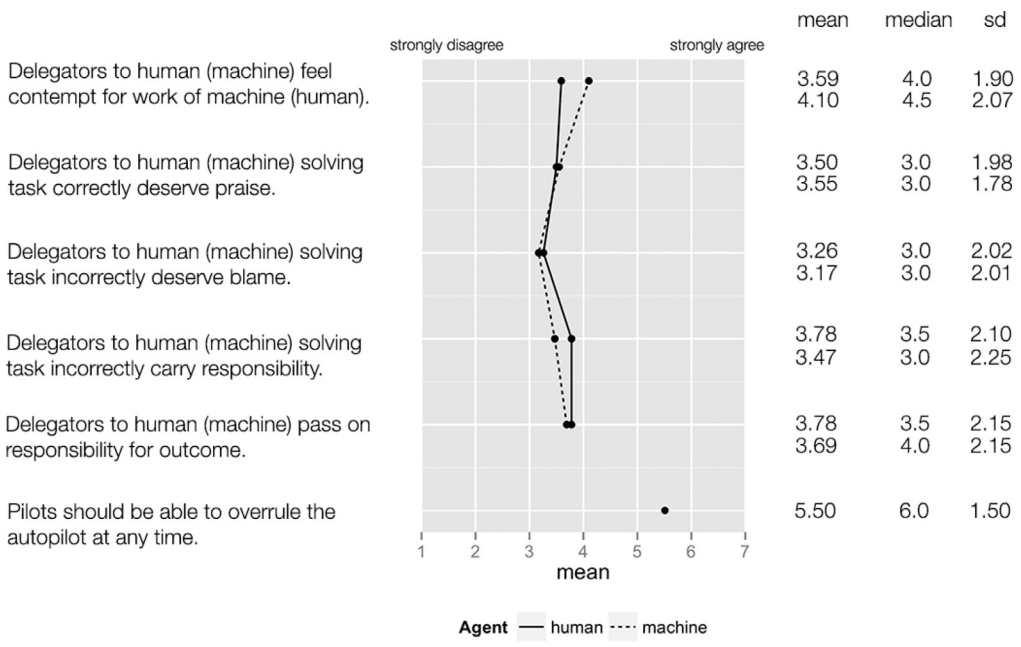


Fig. 3. Results of post-experimental survey.
NOTE: Upper rows of numbers to the right of the graph represent means, medians and standard deviations (sd) for human agent, lower rows represent measures for machine agent.

toward humans and machines between subjects who made different delegation decisions. In particular, one might expect that delegators to humans are generally more skeptical toward machines and express a lower level of trust. Subjects who delegated to a human task-solver in the first part, however, did not, on average, transfer any less to a machine than to a human (31.63 ECU ($sd = 15.46$) vs. 32.92 ECU ($sd = 16.37$), $p = .627$ according to two-sided Mann–Whitney U-tests). Therefore, our fourth hypothesis is also rejected.

Result 4. The level of trust toward machines and toward humans does not differ significantly.

4. Post-experimental survey

Because both potential explanations for the very clear relative aversion to machine use in the moral domain could not be supported in the experiment, we conducted a survey study in February 2018. In this survey, we recruited 78 new participants via ORSEE (Greiner, 2004) and confronted them with a concise description of part 1 of the experiment. They were then asked to indicate their agreement to several statements on a 7-point Likert scale.

We confronted subjects with five pairs of statements that were identical except for the words “human task-solver” and “machine”. The order in which statements were presented was randomized within each pair. Specifically, we investigated the following alternative explanations for the observed aversion to a delegation to a machine. First, people might feel that a delegator who delegated the task to a machine holds a human task-solver’s benevolent effort in contempt (first pair of statements). Second, people might be biased against delegators to machines when attributing praise and blame for a resulting outcome (second and third pair). Third, delegators might successfully pass on the responsibility for negative outcomes to another human but not to a machine (fourth and fifth pair). The eleventh statement was not directly related to part 1 but represented a general remark on automation in a morally relevant domain that we included as a control.

Fig. 3 illustrates the average agreement to the five pairs of statements testing the alternative explanations for an aversion to machine use in the moral domain and to the eleventh statement that served as a control.

The figure suggests that the agreements for each pair of statements concerning delegators to a human and to a machine are quite similar to each other. In fact, none of the differences between the delegators to humans and machines is significant ($p > .100$ according to two-sided Wilcoxon signed-rank tests) except for the first pair. People do indeed more readily agree with the idea that delegators to machines hold a human’s work in contempt than that delegators to humans hold the machine’s work in contempt ($p = .031$). The agreement with the former statement, however, is still close to neutrality. The effect is thus likely to be driven by a perceived implausibility of the idea that a machine’s effort can be held in contempt.

The only statement where participants’ answers tend to clearly deviate from neutrality is the eleventh statement, where people firmly express the opinion that a human pilot should always be able to overrule the decision of an autopilot. While we can thus also identify an aversion to automation in the moral domain in the post-experimental survey, none of the alternative explanations tested in this survey can be supported.

5. Discussion

In this study, we compared the frequency of delegation decisions of a task that affects a third party to machines and humans and elicited their respective evaluation by impartial observers. It should be stressed again that the question we posed here was about people’s preference relation over a machine agent and a human agent, and not about delegating versus performing the task oneself. Consequently, subjects had to delegate in either case and could thus not be blamed merely for shifting responsibility.

We found that subjects express an aversion to delegating tasks that fall into the moral domain to machines rather than humans. First, this manifests in the relatively small fraction of delegators that mandate a machine rather than a human. Second, it is clear that observers evaluate the decision to delegate to a machine in the moral domain less favorably than if the delegation is to a human. Interestingly, machine use is

viewed more critically, irrespective of whether the delegation ultimately caused positive or negative consequences for the person affected.

The experiment tested two potential explanations for an aversion to machine use in the moral domain: an oversensitivity to machine errors, and a lack of trust in machines. Both explanations could be ruled out in our experiment. Subjects did not perceive machine errors more saliently than human errors, as subjects' incentivized guessing of failure rates demonstrates. The phenomenon identified, therefore, seems to be an aversion to delegating tasks that affect a third party to machines *per se* as opposed to an instrumentally justified attempt to minimize the risk of failure for those affected. Analogously, the level of trust expressed by subjects toward a machine agent was very similar to the trust level expressed toward a human. Thus, we were unable to identify a general distrust in machines in a self-regarding trust game. This latter finding indicates that the unconditional aversion to machine use seems to be rather specific to the delegation of tasks that affect a third party.

Finally, we used a post-experimental survey with fresh subjects to investigate three alternative explanations for the observed phenomenon: the feeling that delegators to machines hold humans' effort in contempt, a bias against delegators to machines when attributing praise and blame, and a difference in a delegator's ability to pass on responsibility to another human and to a machine. Neither of these potential alternative explanations, however, could account for the aversion at hand.

From our findings, it seems that most people rather intuitively dislike machine use in the moral domain—an intuition which turns out being hard to rationalize. We identified this aversion per se by experimentally equalizing humans' and the algorithm's performance. In practice, however, algorithms will usually not be simulating human moral behavior but will be programmed to implement a specific normative rationale. This attachment to rules may induce a dismissal of their decisional inflexibility in people. Such an instrumental aversion would then come in addition to the non-instrumental aversion that we identified. In the case of self-learning algorithms, we might observe an additional instrumental aversion to decisional opacity.

Our results underline the importance of an open discussion of machine use in the moral domain. The case of automated driving certainly qualifies as such a domain, since errors of the machine may cause substantial externalities to third parties. The non-instrumental aversion identified suggests that the emphasis on the superior performance of automated cars, which is currently the main argument for automation in traffic, may not be sufficient or even decisive in convincing the general public. It might be as important to address the perceived moral problems that are necessarily associated with the introduction of automated vehicles.

Against this background, Chris Urmson, head of Google's self-driving car project, might be mistaken in downplaying the role of moral considerations in the context of automated driving by calling them "a fun problem for philosophers to think about" (McFarland, 2015). As this empirical study suggests, concerns regarding the involvement of machines in the moral domain are not only an issue for armchair philosophers but may reflect a larger societal phenomenon, viz. a folk aversion (see also Kohl et al. (2018)). So far, the industry seems to mainly be occupied with engineering issues and has, due to a déformation professionnelle, predominantly neglected or downplayed the possibility of public resistance to the new technology. It may, however, be well-advised to take moral concerns against automated driving seriously, since citizens' resistance may slow down the automation process substantially. This, however, would mean to preserve a status quo that involves an avoidably high number of traffic deaths, injuries and damages.

Research that investigates how the feeling of unease can be addressed prophylactically (Feldhütter et al., 2016) is just emerging. Enabling people to experience, and thus better understand, the technology in order to dissipate reservations and fears may pave the way for a trouble-free introduction of autonomous driving. A deeper investigation of the causes of people's aversion to the use of automated cars in the moral domain seems to us a promising venue for future research.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:http://dx.doi.org/10.1016/j.socec.2018.04.003.

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